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MILITARY SPECIFICATION

CARTRIDGE, 57MM, HE, M306A1, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of high explosive cartridge for use in 57MM rifles M18 and M18A1, designated as Cartridge, 57MM, HE, M306A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — General Specification for Ammunition except Small Arms Ammunition.

MIL-R-11470 — Radiographic Inspection; Qualification of Equipment, Operators and Procedures.

MIL-R-11471 — Radiographic Inspection of Metals.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

MIL-STD-1168 — Lot Numbering of Ammunition.

MIL-STD-1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ARMY

9215029 — Projectile, 57MM, HE, M306A1, Loading Assembly.

FSC 1310

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9215030 — Cartridge, 57MM, HE, M306A1, W/F PD, M503A2, for 57MM Rifle, M18 and M18A1.

9215018 — Ammunition Container, Fiber, 57 MM, M167A2, Assembly, details, packing and marking.

8796471 — Box, Packing, Ammunition, for 57MM Rifle Ammunition, in Fiber container, M167A2, Assembly, details and packing.

PUBLICATIONS

ARMY

ET–9215030 — Equipment Tabulation.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material and parts shall be in accordance with applicable drawings and specifications.

3.1. *Clean explosive scrap.* Clean explosive scrap produced in regular operations may be remelted and reused provided that it complies with the requirements for the grade of high explosive specified, except for form. Scrap from floor sweeping, ventilation and suction apparatus (other than from drilling operations specifically controlled to prevent contamination of explosive) shall not be used (see 6.7).

3.2 Cartridges. The cartridges shall comply with all requirements specified on Drawing (dwg.) 9215030 and with all requirements specified in applicable specifications.

3.3 Cavitation in high explosive charge.

3.3.1 *Projected cavities.* Cavities having a projected length of 1/32 inch (in.) or less shall be disregarded. Cavities within the explosive charge shall not exceed the requirements specified in table I, when determined as specified in 4.3.1 (see figure 1).

TABLE I

	A	B
Sum of projected areas of the cavities, excluding pipes, cracks, and annular rings, square inch (sq. in.)	*1/64	1/2
Projected length of any cavity, excluding pipes, cracks and annular rings (in.)	1/8	1/2
Piping cavities, maximum (max.) projected area (sq. in.)	0	1/2
Piping cavities, max, projected width (in.)	0	1/4
Cracks, max, projected width, (in.)	1/32	1/32
Annular rings, max, projected width (in.)	0	1/4

* If the length of the largest cavity is 1/16 inch or less, the maximum total projected area may be 1/20 square inch.

3.3.2 *Porous areas.* Porous areas shall be treated as cavities and shall be subjected to the restrictions placed upon cavities, except that 80 percent of the projected length and 80 percent of the projected area shall be considered for acceptance purposes.

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3.3.3 *Cracked charges.* Not more than two transverse cracks shall be permitted in any charge, and not more than one transverse crack shall be permitted in Segment A.

3.3.4 *Base separation.* There shall be no separation between the shell base and explosive charge.

3.3.5 *Surface Cavitation.* Cavities in the base of the fuze well shall not be cause for rejection provided the sum of the areas of the cavities is not greater than 80 percent of the area of the fuze well base and no individual point extends more than ¼ inch below the maximum depth of the fuze well permitted by the drawing.

3.4 Crimp.

3.4.1 *Torque.* The shell shall withstand a torque of 100 inch-pounds minimum, without rotating in the cartridge case when tested as specified in 4.3.2.1.

3.4.2 *Bullet pull.* The cartridge case shall separate from shell at a load between 1,000 pounds minimum and 2,500 pounds maximum, when tested as specified in 4.3.2.2.

3.5 Proving ground. The cartridge shall comply with the following requirements when tested as specified in 4.3.3.

3.5.1 *Muzzle velocity.* The corrected muzzle velocity of the projectile at ambient temperature shall be not more than 1236 feet per second (fps) nor less than 1164 fps and the standard deviation of the lot shall be no greater than 13 fps (see 4.3.3.1).

3.5.2 *Projectile functioning.* The projectile shall not fail to function with high order detonation.

3.5.3 *Premature bursts and security of parts.* There shall be no premature burst or indication of metal parts failure in the gun bore or in flight before impact.

3.5.4 *Safety.* There shall be no premature burst or indication of metal parts separation in the gun bore or in flight when fired with propellant adjusted to give 112 percent of the rated maximum pressure for the cartridge.

3.6 Propellant weight. The weight of the propellant shall not differ by more than plus or minus 0.02 ounce from that designated by the loading authorization for the particular lot of propellant used (see 4.2.3.4).

3.7 Workmanship.

3.7.1 *Cartridge.* All manufacturing processing and assembly operations of the cartridge shall be correctly performed in a thoroughly workmanlike manner. Parts shall be free of burrs, chips, dirt, cracks, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.7.2 *Loading of high explosive.* The interior surface of the projectile shall be dry when the high explosive is poured into the projectile. Solid explosives shall not be present in the projectile before loading or be added to the projectile during pouring or cooling.

3.7.3 *Painting and marking.* The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked when such painting and marking has been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is respon-

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sible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL–STD–109 in order to define terms used herein. The provisions of MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all of the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspections performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

(e) Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractural requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance or interchangeability as determined by the Government shall not necessitate changing the lot interfix number. The examinations and tests listed herein shall be performed on inspection lots submitted in accordance with MIL–STD–105, MIL–STD–1235, or MIL–STD–414, as applicable. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL–STD–1168. Each lot shall contain—

(a) Cartridge cases of one lot interfix number from one manufacturer.

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(b) Projectile metal parts of one lot interfix number from one manufacturer.

(c) Propellant from not more than lot.

(d) Loaded primers of one lot interfix number from one manufacturer.

(e) Liners from one manufacturer produced in accordance with the same drawing, same drawing revision, same specification, and same specification revision.

(f) Loaded fuzes of one lot interfix number from one manufacturer.

(g) High explosive from one lot interfix number from one manufacturer.

4.2.2 *Examination.* Inspection for Critical defects (and Major defects, when so specified), shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Projectile loading assembly, prior to loading explosive (see dwg. 9215029, covering a detail of dwg. 9215030).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical:			
1.	Foreign matter in charge cavity	Visual	01001
2.	Unsolidified pool of paint in bottom of cavity	Visual	01002
Major:	AQL 0.40 percent		
101.	Interior protective coating with bare spot or blister	Visual	01003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	01004

4.2.2.2 Projectile loading assembly, prior to assembly of fuze (see dwg. 9215029 covering a detail of 9215030).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Depth of fuze well, maximum, (max.)	Gage	02001
Minor:	AQL 1.5 percent		
201.	Diameter of fuze well, max.	Gage	02002
202.	Depth to explosive charge	Gage	02003

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Categories	*Defects*	*Method of inspection*	*Code No.*
203.	Evidence of poor workmanship (see 3.7)	Visual	02004
204.	Surface cavitation excessive (see 3.3.5)	Visual	02005

4.2.2.3 Cartridge case of primer prior to assembly (see dwg. 8839466 covering a detail of dwg. 9215030).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	One or more flash holes missing	Visual	03001
Major:	AQL 0.40 percent		
101.	Paper liner in flash tube damaged	Visual	03002
Minor:	AQL 1.0 percent		
201.	Evidence of poor workmanship		

4.2.2.4 Cartridge case, prior to loading propellant (see dwg. 9215030).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Primer obviously loose	Manual	04001
Major:	AQL 0.40 percent		
101.	Primer damaged	Visual	04002
Minor:	AQL 1.5 percent		
201.	Foreign matter on interior of liner	Visual	04003
202.	Foreign matter in charge	Visual	04004
203.	Evidence of poor workmanship (see 3.7)	Visual	04005

4.2.2.5 Cartridge case with propelling charge, liner closed and sealed (see dwg. 9215030).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Liner damaged to extent that propellant can escape	Visual	05001
Major:	AQL 0.65 percent		
101.	Propellant charge missing	Visual	05002
102.	Liner improperly closed	Visual	05003

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Categories	*Defects*	*Method of inspection*	*Code No.*
103.	Liner damaged to extent that moisture barrier is broken but not to the extent that propellant can escape	Visual	05004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	05005

4.2.2.6 Cartridge assembly, prior to assembling of fuze (see dwg. 9215030).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Foreign matter in fuze well	Visual	06001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	06002

4.2.2.7 Cartridge assembly (see dwg. 9215030).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
1.	Primer above flush	Gage	07001
Major:	AQL 1.0 percent		
101.	Torque test failure of fuze	Test	07002
102.	Fuze not fully seated	Visual	07003
103.	Rotating band damaged	Visual	07004
104.	Primer percussion element below flush, max.	Gage	07005
Minor:	AQL 1.0 percent		
201.	Total length, max.	Gage	07006
202.	Marking missing, misleading or unidentifiable, including correct color	Visual	07007
203.	Evidence of poor workmanship (see 3.7)	Visual	07008

4.2.2.8 Unsealed fiber container (see dwg. 9215018).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Packing component missing	Visual	08001

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Categories	*Defects*	*Method of inspection*	*Code No.*
Minor:	AQL 1.0 percent		
201.	Round excessively tight in container	Manual	08002
202.	Evidence of poor workmanship (see 3.7)	Visual	08003

4.2.2.9 Sealed fiber container (see dwg. 9215018).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Container cut or damaged through all asphalt impregnated layers	Visual	09001
102.	Tape incomplete or badly wrinkled	Visual	09002
103.	Metal end loose or distorted	Visual	09003
Minor:	AQL 2.5 percent		
201.	Tear tab missing or improper	Visual	09004
202.	Gap between cover and body of container	Visual	09005
203.	Color of tape incorrect	Visual	09006
204.	Marking missing, misleading or unidentifiable	Visual	09007
205.	Contents loose	Manual	09008

4.2.2.10 Sealed wood packing box (see dwg. 8796471).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Box damaged	Visual	10001
102.	Hardware or strapping missing, broken or loose	Visual-Manual	10002
103.	Boards loose or nails protruding	Visual	10003
Minor:	AQL 2.5 percent		
201.	Hardware improperly engaged	Visual-Manual	10004
202.	Strapping improperly assembled	Visual-Manual	10005
203.	Car seal missing, unsealed or improperly positioned	Visual-Manual	10006
204.	Contents loose	Manual	10007
205.	Handle missing or insecure	Visual	10008
206.	Marking missing, misleading or unidentifiable	Visual	10009

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4.2.3 Testing.

4.2.3.1 *Check test for possible deterioration of the primer—Major defect—Code No. 02005.* If the total elapsed time between original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds 2 years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in the applicable primer loading specification immediately before the primer lot is assembled into the cartridges. The test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the cartridges (see 6.4).

4.2.3.2 *Cavitation (see 3.3 except 3.3.5) (see dwg. 9215029)—Critical defect—Code No. 01005.* The sampling plan and procedures for this test shall be in accordance with MIL–STD–1235, using a CSP–1 plan having an AQL of 0.015 percent. Appropriate f and i values shall be as specified by the procuring activity. Any cartridge having cavitation in excess of that permitted by the applicable requirement for any segment shall be classed defective. The test shall be performed as specified in 4.3.1, using equipment in accordance with 4.2.4.

4.2.3.3 *Profile and alignment (see dwg. 9215030)—Major defect—Code No. 07015.* The cartridges, including those which have had the bourrelet repainted (see 3.7.3) shall be gaged 100 percent for profile and alignment. Any cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.2.4.

4.2.3.4 *Propellant weight (see 3.6)—Critical defect—Code No. 04005.* The propellant shall be weighed and check weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing and, if performed manually, shall be performed by another operator. Any charge which fails to comply with the applicable requirement shall be classed defective and removed from the lot.

4.2.3.5 Crimp.

4.2.3.5.1 *Torque (see 3.4.1)—Major defect—Code No. 07016.* The sampling plan for this test shall be in accordance with MIL–STD–105, and an AQL of 0.40 percent.

4.2.3.5.2 *Bullet Pull (see 3.4.2)—Major defect—Code No. 07017.* The sampling plan for this test shall be in accordance with MIL–STD–414, Section B, Part I, Table B1, Code Letter F, and AQL of 1.5 percent.

4.2.3.6 *Proving ground.* The following tests shall be performed as specified in 4.3.3.

4.2.3.6.1 *Muzzle velocity (see 3.5.1)—Major defect—Code No. 07011.*

4.2.3.6.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL–STD–414, Section B, Table B-3, Code Letter I and AQL of 1.0 percent. In addition, if the standard deviation of the sample projectile multiplied by the factor 0.75 exceeds the applicable requirement, the lot shall be rejected.

4.2.3.6.1.2 After the three consecutive lots have met the criteria of 4.2.3.6.1.1, the sampling plan and procedure for this test shall be in accordance with MIL–STD–414, Section B, Table B-3, Code Letter F, and AQL of 1.0 percent. In addition, if the standard deviation of the sample projectile multiplied by the factor 0.65 exceeds the applicable lot requirement, the lot shall be rejected.

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4.2.3.6.2 *Projectile functioning (see 3.5.2) —Major defect—Code No. 07012.*

4.2.3.6.2.1 Beginning with the first lot and continuing until three consecutive lots have complied with the applicable requirements specified, the cartridges fired for muzzle velocity in 4.2.3.6.1.1 shall be observed for functioning. If 3 or more projectiles fail to comply with the applicable requirement, the lot shall be rejected.

4.2.3.6.2.2 After the three consecutive lots have met the criteria of 4.2.3.6.2.1, the cartridges fired for muzzle velocity in 4.2.3.6.1.2 shall be observed for functioning. If 2 or more projectiles fail to comply with the applicable requirement, the lot shall be rejected.

4.2.3.6.3 *Premature bursts and security of parts (see 3.5.3)—Critical defect—Code No. 07013.* The sample cartridges fired in the muzzle velocity shall be observed for this test. If one or more rounds fail to comply with the applicable requirement, the lot shall be rejected.

4.2.3.6.4 *Safety (see 3.5.4)—Critical defect—Code No. 07014.*

4.2.3.6.4.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, 15 sample projectiles shall be selected at random from each lot for this test. If any projectile fails to comply with the safety requirements, the lot shall be rejected.

4.2.3.6.4.2 After three consecutive lots have met the criteria of 4.2.3.6.4.1, 5 sample projectiles shall be selected at random from each lot for this test. If any projectile fails to comply with the safety requirements, the lot shall be rejected.

4.2.4 *Inspection Equipment.* Equipment Tabulation Number ET–9215030 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection Equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the ET shall be in accordance with MIL–I–45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Determination of cavities.* The sample cartridges shall be subjected to x-ray examination. In all exposures the x-ray beam shall form a 90 degree angle with the longitudinal axis of the projectile. Examination of the x-ray negative shall be made to determine compliance with applicable requirements. (Note: Radiographic equipment, operations and procedures shall be qualified in accordance with MIL–R–11470). In addition, the sensitivity of x-ray techniques shall be 2.0 percent, and the photographic density of the film used shall

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be no less than 1.25, as determined in accordance with 4.3.1.2. In the event an original x-ray picture leaves doubt as to acceptability, the cartridge shall be re-x-rayed at an angle 90 degrees from that of the original exposure and in the same plane.

4.3.1.1 *Check of sensitivity of x-ray technique.* The sensitivity of x-ray technique shall be determined by the means of a penetrameter fabricated as specified herein. The penetrameter shall be placed on the side of the projectile away from the film (source side). Observation shall be made of the image of the penetrameter on the x-ray film. The sensitivity of x-ray technique shall be considered satisfactory when the image of the penetrameter including the hole "T2", is sharply defined on the x-ray film. Unless otherwise specified, the penetrameter shall be made of steel having an approximate specific gravity of 7.8. The thickness of the penetrameter shall be based on a combination of both charge and steel thicknesses and shall be equal to 2.0 percent of the max. equivalent thickness. The penetrameter shall have a hole the diameter of which is twice the thickness of the penetrameter. For further details on construction of the penetrameter see MIL–R–11471. The actual thickness may be determined from a radiograph of the metal parts. The penetrameter thickness shall be determined as illustrated by figure 2 of this specification.

4.3.1.2 *Check of photographic density of film.* The photographic density of the film shall be determined on the film in a region of sound charge plus metal corresponding to that portion of the material which is of the greatest combined thickness to be examined on the negative. To judge densities, it is recommended that standards be used (see 6.8). Comparison of film with standards may be made visually; however, the use of a densitometer is recommended.

4.3.2 Crimp.

4.3.2.1 *Torque.* The security of the crimp of the cartridge case to the projectile shall be determined by applying a torque load at a rate of travel not to exceed 10 degrees per minute. Observation shall be made for compliance with the applicable requirement.

4.3.2.2 *Bullet pull.* The security of the crimp of the cartridge case to the projectile shall be determined by applying a load in the axial direction with a rate of travel of the testing machine head of 0.100 to 0.125 inch per minute. Observation shall be made for compliance with the applicable requirement.

4.3.3 *Proving ground.* These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired in a gun for which the ammunition is standard. Observation shall be made for conformance to all requirements.

4.3.3.1 *Muzzle velocity.* Ten of the 25 sample cartridges tested for muzzle velocity from each of the first 3 lots shall be fired from a ballistic penduium and the muzzle velocities and recoil momentum shall be reported for each round. After the first 3 lots the use of the ballistic pendulum is not required. Live fuzes shall not be fired through velocity screens other than the solenoid type.

4.3.3.2 *Projectile functioning.*

4.3.3.2.1 *Vertical target.* Fifteen cartridges of those fired from each of the first three lots and five cartridges of those fired from each subsequent lot for the velocity test shall impact against a 1-inch wood bursting screen placed not less than 400 feet from the muzzle of the rifle. The cartridges shall be observed for functioning.

4.3.3.2.2 *Horizontal target.* Ten cartridges of those fired from each of the first three lots and five cartridges of those fired from

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each subsequent lot for the velocity test shall impact on firm ground with the rifle elevated at not less than 320 mils. The cartridges shall be observed for functioning.

4.3.3.3 *Premature bursts and security of parts.* All cartridges fired in the proving ground tests shall be observed for premature bursts, parts failures, and other apparent safety hazards.

4.3.3.4 *Safety.* The sample cartridges shall be assembled with dummy fuzes and fired with propelling charges adjusted to give 112 percent of the rated maximum pressure for the cartridge. In addition, for the first three production lots from each manufacturer (loading facility), a smear camera shall be used to observe the projectile in flight, in order to record any possible malfunctioning.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 *Level A.* The cartridges shall be preserved and packaged in accordance with dwg. 9215018.

5.2 Packing

5.2.1 *Level A.* The cartridges (in fiber container) shall be packed in accordance with dwg. 8796471.

5.3 Marking. Marking of the wood box shall be in accordance with dwg. 8796471. The fiber container shall be marked in accordance with dwg. 9215018.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Ammunition data cards shall be prepared for each lot in accordance with MIL–STD–1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (SMUPA Form 1010) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the EL by the following codes:

CDAF — Contractor's design responsibility on Army format in accordance with MIL–D–45608.

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CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibrating instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format

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and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Unless otherwise specified on the Equipment List, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Cost of check test. The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Submission of test data. In addition to the normal distribution of records, when the cartridge is procured by the Department of the Army, one copy of all test data shall be forwarded to each of the following:

Commanding Officer.

Picatinny Arsenal

ATTN: SMUPA–ND2 and SMUPA–DC1

Dover, New Jersey

6.6 Supersession data. This specification supersedes in part MIL–C–1391A.

6.7 Clean explosive scrap. Clean explosive scrap is defined as scrap formed in kettles, pails, risers and loading machines used in the pouring and cooling operations, and scrap formed during the drilling operation.

6.8 Photographic densities. "American Standard for Diffuse Transmission Density" published by American Standards Association, 70 East 40th Street, New York, N. Y., has been found satisfactory as a source of information of filter densities.

Custodian:

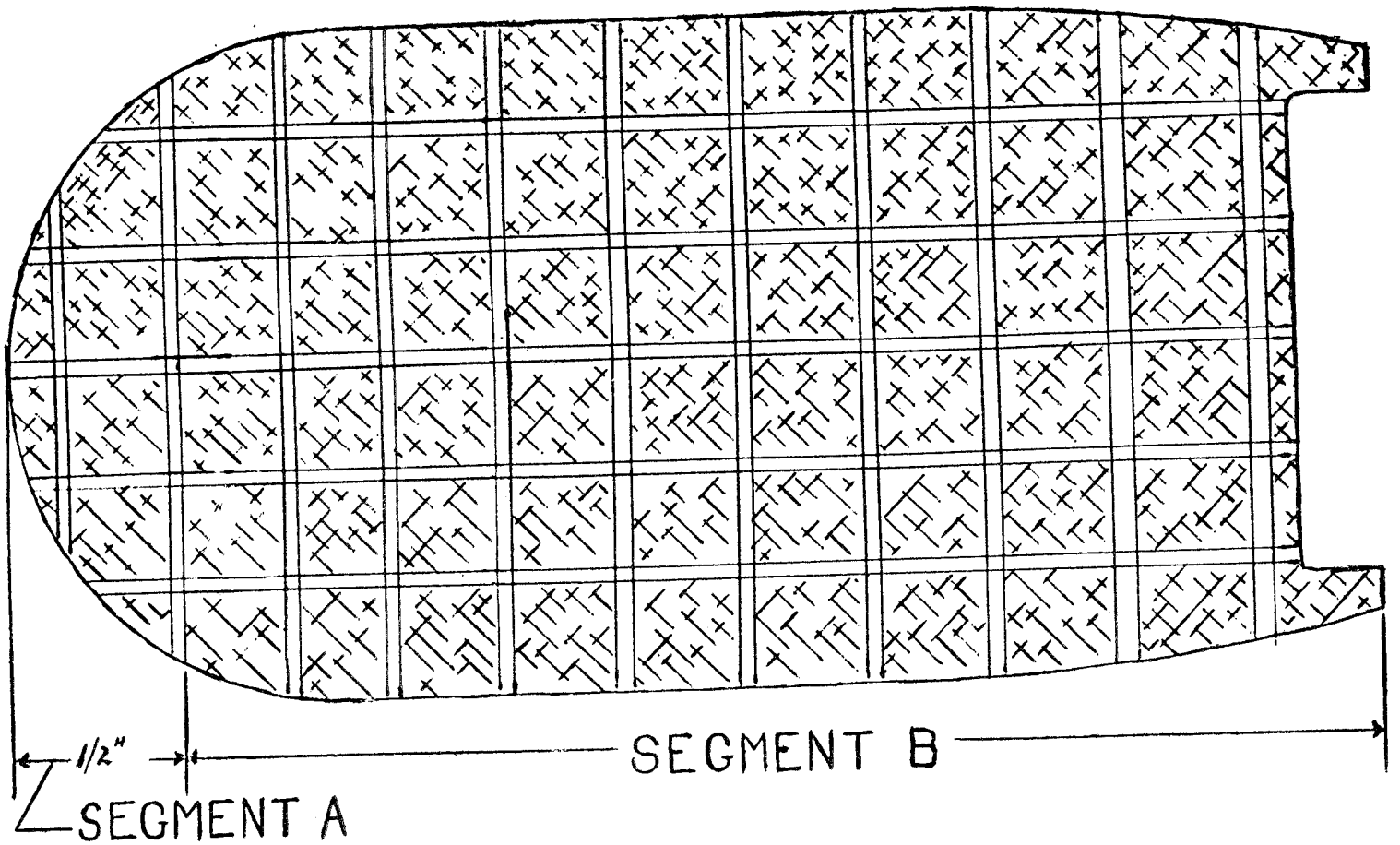
Army—MU

Preparing activity:

Army—MU

Project No. 1310–A030

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 OCT 64 1426

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